

JIAYI WANG

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HIGHLIGHTS OF RESEARCH & INTERNSHIP INTERESTS

- **Research highlights & interests:** Efficient GPGPU architectures, reconfigurable architectures, memory architectures, and AI accelerators.
- **End-to-end digital ASIC tape-out:** RTL design and verification, synthesis, physical design, static timing analysis, and physical signoff (DRC/LVS/PEX).
- **Leadership & mentoring:** Lead Teaching Assistant (TA) for University of Washington’s digital tape-out courses; mentored 12 TSMC 180 nm tape-out projects, guiding RTL design and verification through physical design and tape-out.

EDUCATION

University of Washington, Seattle Ph.D. in Computer Engineering GPA: 4.00 Focus: Computer architecture, reconfigurable AI hardware architecture, AI systems, digital VLSI Advisor: Prof. Ang Li	Sep 2023 – June 2028 (expected)
University of Washington, Seattle M.S. in Electrical Engineering GPA: 4.00 Focus: Computer architecture, digital VLSI	Sep 2020 – Aug 2022
Shanghai Jiao Tong University B.S. in Electrical Engineering GPA: 3.70 Focus: Power electronics, embedded systems	Sep 2016 – Jun 2020
University of California, Los Angeles Summer School, Computer Science GPA: 4.00	Summer 2019

FIRST-AUTHOR PUBLICATIONS AND WORKSHOPS

- [1] **Jiayi Wang**, Ang Da Lu, Zhichen Zeng, and Ang Li,
“*DICE: Enabling Efficient General-Purpose SIMT Execution with Statically Scheduled Coarse-Grained Reconfigurable Arrays*,”
in **The 53rd Annual International Symposium on Computer Architecture (ISCA 2026)**, Raleigh, NC, USA, Jun. 27–Jul. 1, 2026. (Acceptance rate: 19.1%) [[arXiv:2605.05496](#)] [[Slides](#)]
- [2] **Jiayi Wang**, Kearnan Bishop, and Ang Li,
“*MFSA: A Multi-Format Systolic Array with Native Micro Scaling Support*,”
in **The 44th IEEE International Conference on Computer Design (ICCD 2026)**, Hong Kong, Nov. 16–18, 2026. (Acceptance rate: 26%.)
- [3] **Jiayi Wang**, Maohua Nie, Sin-Chen Lin, C.-J. Richard Shi, and Ang Li,
“*TransDot: An Area-Efficient Reconfigurable Floating-Point Unit for Trans-Precision Dot-Product Accumulation for FPGA AI Engines*,”
in **The 34th IEEE International Symposium on Field-Programmable Custom Computing Machines (FCCM 2026)**, Atlanta, GA, USA, May 13–16, 2026. (Acceptance rate: 24.6%.) [[IEEE Xplore](#)]

- [4] **Jiayi Wang**, Ang Da Lu, and Ang Li,
“*DICE: Efficient Thread-Pipelined SIMT Execution on a Statically Scheduled Reconfigurable Spatial Fabric*,”
Presented at 18th Workshop on General Purpose Processing Using GPU (GPGPU 2026), held
in conjunction with ASPLOS 2026, Pittsburgh, PA, USA, Mar. 22–23, 2026.
- [5] **Jiayi Wang**, Chenyi Wang, and Ang Li,
“*piPE-SA: Enabling Deeply Pipelined Processing Elements in Systolic Arrays*,”
in **Energy Efficient Machine Learning and Cognitive Computing (EMC2-11) Workshop**, held in
conjunction with ASPLOS 2026, Pittsburgh, PA, USA, Mar. 22, 2026.
- [6] **Jiayi Wang**, Yiwei Yang, Jiajin Wu, Zheng Zeng, Di Lu, and Lian Lian,
“*Hybrid Aerial-Aquatic Vehicle for Large-Scale High-Spatial-Resolution Marine Observation*,”
in **OCEANS 2019 Marseille**, Jun. 2019, pp. 1–7. doi: [10.1109/OCEANSE.2019.8867402](https://doi.org/10.1109/OCEANSE.2019.8867402).

COLLABORATIVE PUBLICATIONS

- [7] Zhichen Zeng, Xichong Zhang, Junpan Wu, Yifei Zuo, Chi-Chih Chang, **Jiayi Wang**, Maohua Nie, Ji Liu, Ang Li, and Banghua Zhu,
“*Step-dLLM: Adaptive Step-aware Sparse Attention for Efficient Diffusion LLM Inference*,”
Conference on Neural Information Processing Systems (NeurIPS 2026). Accepted.
- [8] Maohua Nie, Jiang Zhu, Jingqun Zhang, Zhichen Zeng, **Jiayi Wang**, Sibozhang, Jialin Wang, and C.-J. Richard Shi,
“*HierSVA: A Data Synthesis Pipeline, Dataset, and Benchmark for LLM-Driven Hierarchical Hardware Formal Verification*,”
Conference on Neural Information Processing Systems (NeurIPS 2026). Accepted. [\[arXiv\]](#)
- [9] Jingqun Zhang, Maohua Nie, **Jiayi Wang**, Weihang Li, Rishi Sappidi, Shwet Chitnis, and Ang Li,
“*CacheFlex: Direct Software-Managed Access to Higher-Level Cache for Scalable Vector Support*,”
in **The 59th IEEE/ACM International Symposium on Microarchitecture (MICRO 2026)**. (Acceptance rate: 25.3%.)
- [10] Ryan H. Lee, Kate L. Tseng, Te Min Yu, Andrew Pan, **Jiayi Wang**, James Rosenthal, Kevin J. Ho, Ang Li, and Matthew Reynolds,
“*All-Digital Bluetooth Low Energy (BLE) Backscatter ASIC Using Standard I/O Pad Drivers in 180 nm CMOS*,”
in **Proceedings of the 2026 IEEE International Conference on RFID (RFID 2026)**.
- [11] Jingqun Zhang, Weihang Li, Maohua Nie, Yung-Jen Cheng, **Jiayi Wang**, Shwet Chitnis, and Ang Li,
“*CacheFlex: Explicitly Control What You Need in Your Cache*,”
Presented at Energy Efficient Machine Learning and Cognitive Computing (EMC2-11) Workshop, held in conjunction with ASPLOS 2026, Pittsburgh, PA, USA, Mar. 22, 2026.
- [12] Ang Da Lu, **Jiayi Wang**, and Ang Li,
“*STEP: Spatially Threaded Execution Pipeline*,”
in **Proceedings of the 6th Workshop on Languages, Tools, and Techniques for Accelerator Design (LATTE 2026)**, held in conjunction with ASPLOS 2026, Pittsburgh, PA, USA, Mar. 22–23, 2026. [\[PDF\]](#)
- [13] Shwet Chitnis, Fergus Xu, Ayush Kulkarni, **Jiayi Wang**, Jingqun Zhang, Arjun Raje, and Ang Li,
“*Precision-Aware Communication in CGRAs*,”
in **The 34th IEEE International Symposium on Field-Programmable Custom Computing Machines (FCCM 2026)**.
- [14] Shwet Chitnis, **Jiayi Wang**, Fergus Xu, Ayush Kulkarni, Rampranav Navendran, Juwon Jun, Jingqun Zhang, and Ang Li,
“*DORA: Open-Source Infrastructure for Prototyping Reconfigurable Fabrics*,”
Poster at Workshop on Open-Source Computer Architecture Research (OSCAR 2026), held in conjunction with ISCA 2026, Raleigh, NC, USA, Jun. 2026.

PATENTS

- [1] Ang Li and Jiayi Wang,
“DICE: Enabling Efficient General-Purpose SIMT Execution with Coarse-Grained Reconfigurable Arrays,”
U.S. Provisional Patent Application 64/012,051, filed Mar. 20, 2026.
- [2] Zheng Zeng, Jiayi Wang, Yiwei Yang, Jiajin Wu, Lian Lian, and Di Lu,
“An Aerial-Aquatic Vehicle Combining a Fixed-Wing Aircraft and a Glider,”
Chinese Invention Patent CN 109204812 B, granted Nov. 17, 2020.

SELECTED RESEARCH EXPERIENCE

Reconfigurable AI Hardware and Accelerator Microarchitecture Research

UW P^N Computer Engineering Lab (PI: Prof. Ang Li)

Aug 2025 – Present

- **TransDot:** Developed a unified FPU supporting FP4/FP8/FP16/FP32 fused multiply-add, subword SIMD, and dot products with FP16/FP32 accumulation, replacing separate per-format FPUs in FPGA AI engines.
- **MFSA:** Designed a weight-stationary systolic array with a shared dot-product datapath supporting formats from FP32 to FP4/INT4; introduced online block-scale rebasing to incorporate MX/NVFP4 scales directly into partial sums and reduce block-scaling overhead.
- **piPE-SA:** Developed timing support for multi-stage PEs in weight- and output-stationary arrays, enabling exploration of PE pipeline depth alongside array size and dataflow.
- **Ongoing:** Designing a twisted-torus systolic fabric with an orthogonal wavefront and runtime partitioning into independently scheduled strips, targeting higher utilization on small, non-square GEMMs in LLM serving.
- Related publications: [2], [3], [5]; one manuscript under review.

High-Efficiency CGRA–GPU/CPU Integration Research

UW P^N Computer Engineering Lab (PI: Prof. Ang Li)

Oct 2024 – Present

- **DICE:** Designed a CGRA-based replacement for the GPU SM’s SIMD backend to reduce register-file traffic and instruction-control overhead while preserving the CUDA programming model.
- Developed a Temporal Memory Coalescing Unit for pipelined memory access and swizzled thread-bank mapping to reduce register-file bank conflicts.
- Built a *p-graph* compiler, contributed to STEP for mapping SIMT kernels onto the spatial fabric, and extended GPGPU-Sim/GPUWattch for performance and power evaluation.
- Achieved **1.90× energy efficiency**, **42% lower power**, and **1.16× performance** relative to a modeled NVIDIA Turing SM baseline on GPU-Rodinia. Co-inventor on a U.S. provisional patent application.
- Led an **8-member RTL team** developing the RTL implementation and FPGA prototype for architectural validation.
- **Ongoing:** Developing a core-private SIMT engine integrated with the CPU cache hierarchy and an in-memory thread-launch mechanism for fine-grained irregular kernels without driver launches or explicit data transfers.
- Related publications: [1], [4], [12]; one manuscript under review.

Coarse-Grained Reconfigurable Architecture Framework Research

UW P^N Computer Engineering Lab (PI: Prof. Ang Li)

Mar 2024 – Present

- **DORA:** Built a parameterized Python framework for CGRA RTL generation with configurable processing elements, routing networks, and control logic; used across the lab's CGRA projects for architecture exploration and FPGA prototyping.
 - Investigated how operand precision informs data movement and interconnect design in CGRAs.
 - Related publications: [13], [14].
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GPU Multi-Tenant Power Management Research

UW P^N Computer Engineering Lab (PI: Prof. Ang Li)

2026 – Present

- Characterized power-induced interference on H100/B200-class GPUs, measuring how a co-located workload increases a prioritized tenant's time-to-first-token under a shared power cap.
 - Proposed per-partition power metering and independent per-GPC voltage/frequency control, with a power arbiter that reallocates budgets at sub-millisecond intervals based on tenant priority, execution phase, and unused capacity.
 - Evaluated the design using real LLM traces and a serving simulator with power models calibrated on H100 and B200; restored the prioritized tenant's latency to near its standalone value in simulation.
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AI Agent Development

SABOL Vision Lab, Cultivate Learning (PI: Prof. Gail E. Joseph)

Feb 2026 – Present

- **VISTA-QEI:** Developed a video-analysis agent that scores teacher-child interactions using the QEI framework and generates coaching reports grounded in video evidence.
 - **VISTA-RT:** Designed a real-time coaching agent integrating live audio/video analysis, rolling context memory, and prioritized prompts delivered through wearable interfaces.
 - Implemented working demos and evaluation tools; presented prototypes to collaborators in AI and early-childhood education.
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High-Speed Interface Integration Research

UW Silicon Systems Research Lab (PI: Prof. C.-J. Richard Shi)

Mar 2022 – Dec 2023

- **CPHY:** Led RTL design, mixed-signal integration, and verification of a DDR/UCIe-compatible PHY and LPDDR4/4x memory controller.
 - Performed physical design and analog/mixed-signal integration for a **TSMC N28 tape-out**.
 - Integrated the controller and PHY with a RISC-V CPU and a deep-learning accelerator.
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INDUSTRY EXPERIENCE

NVIDIA Corporation

Compute Performance Intern

Summer 2026

- Working on GPU acceleration of Cadence EDA tools.

Qualcomm Technologies

Jun 2024 – Sep 2024

Research Intern, GPGPU

- Investigated architectural optimizations in commercial GPGPU/NPU platforms to accelerate sparse matrix multiplication.
- Explored low-level sparse implementation flows in TensorRT, PyTorch, and NVIDIA CUTLASS using NVIDIA 2:4 Sparse Tensor Cores.
- Benchmarked and analyzed sparse Tensor Core performance across multiple models, including ResNet, MobileNet, and BERT.

Orora Design Technologies

Oct 2022 – Dec 2023

Technical Staff

- Developed C++ automation tools for automatic SystemVerilog model generation of analog circuits.

Intel Corporation

Mar 2021 – Sep 2021

Technical Enablement Engineer Intern

- Debugged and verified DDR and PCIe high-speed interfaces on IA-32 platform products.

Signify (Philips Lighting)

Aug 2019 – Feb 2020

Hardware Engineer Intern

- Worked on a dual-mode self-adaptive LED/HID driver circuit for the EU market.

TEACHING, MENTORING, AND SERVICE

University of Washington, Seattle

Jan 2024 – Present

Graduate Teaching Assistant

- Courses supported: EE538 Post-Tapeout Chip Test, EE478 Chip Testing, EE478 VLSI Capstone, EE477/525 VLSI-II, EE476 VLSI-I, and EE271 Digital Circuits.
- Lead Teaching Assistant (TA) for University of Washington's digital tape-out courses; mentored 12 TSMC 180 nm tape-out projects.
- Developed a complete ASIC tapeout flow for TSMC 180 nm technology using the Hammer flow with custom TCL plugins.
- Implemented and maintained synthesis (Genus), place-and-route (Innovus), STA (Tempus), DRC/LVS (Calibre), padframe generation, and hierarchical flows with extracted timing models.

University of Washington, Seattle

Winter 2025

M.S. Admissions Triage Committee

- Reviewed master's applicant materials and provided evaluation feedback.

UW Semiconductor Society

Dec 2025 – Present

Technical Member

- Contributed to student community activities related to semiconductor design and systems research.

EXPERTISE AND SKILLS

- **Architecture and Modeling:** Strong background in AI hardware and CPU/GPGPU/FPGA/CGRA architecture; experienced with GPGPU-Sim/GPUWattch (Accel-Sim/AccelWattch) and gem5.
- **ASIC Design:** End-to-end ASIC design, tapeout, and mentoring experience in TSMC 180 nm and N28, including RTL design, verification, synthesis, place-and-route, STA, and signoff (DRC/LVS/PEX).
- **EDA Tools:** Cadence (Xcelium, Genus, Innovus, Tempus), Synopsys, and Mentor/Calibre.
- **Programming:** CUDA, Python, C++, and SystemVerilog; strong debugging skills and experience working with complex codebases.